

Conflict and Peace

Week 4: Engaging in (organized) crime

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September 26th, 2025

What drives people into crime?

We have already highlighted how economic forces motivate people into conflicts

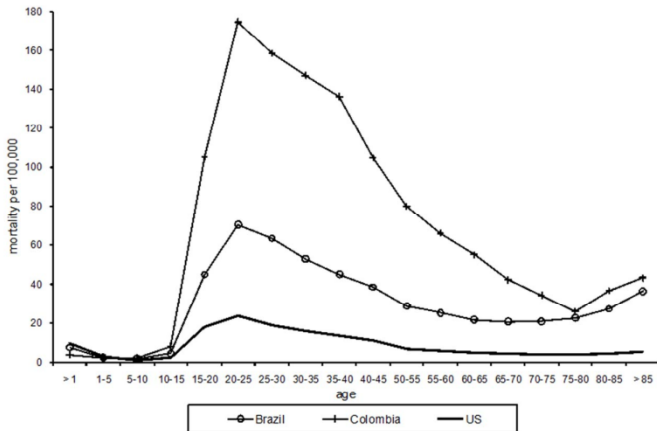
- If opportunity cost of work is higher, people less likely to join crime
 - ▶ Employment prospects, Returns from education, long-run income flow discourage crime
- People often compare their returns in legal markets vs illegal markets
 - ▶ This explains many types of crimes: From joining criminal groups to tax evasion

Here we focus on organized criminal groups

- Not just the violence they cause, but also the strategic motives behind their business
- Also have recruiting mechanisms that target the vulnerable
 - ▶ Youth and the poor, sometimes (ethnically/culturally) marginalized individuals
- Sometimes run their own 'governments' (criminal governance)
 - ▶ Collect "protection fees" (extortions), compete with formal governments for citizen support

Youth disproportionately affected

Figure 1: MORTALITY BY VIOLENCE BY AGE GROUP - SELECTED COUNTRIES
WORLD HEALTH ORGANIZATION DATA (1990s)



Source: Soares, Rodrigo (2006) "The welfare cost of violence across countries", *Journal of Health Economics*, 25, 821-846

Large share of criminal groups, but with short career paths

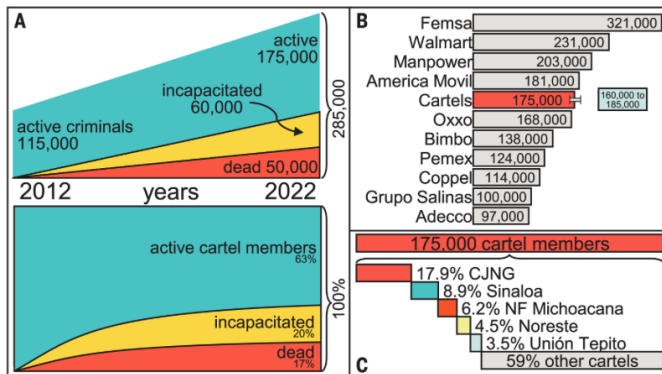


Fig. 3. Current size of cartels and career paths for recruited members.

(A) Between 2012 and 2022, we estimate that 285,000 people took part as cartel members, but only 60% were still active by 2022. The cartel career is brief and risky. Roughly 17% of them are dead, and 20% are incapacitated. (B) Number of employees from the top 10 companies in Mexico and the combined size of cartels (27). We estimate that cartels had between 160,000 and 185,000 members combined. (C) Of the 175,000 active cartel members, roughly 17.9% are part of CJNG, 8.9% are part of Cartel de Sinaloa, and 6.2% are from Nueva Familia Michoacana—the top three cartels in size.

Source: Preito-Curiel et al (2023) "Reducing cartel recruitment is the only way to lower violence in Mexico", *Science*, 381, 1312-1316

Criminal governance is actually widespread

What is criminal governance?

- Provision of order, services, and market regulation by non-state criminal entities
- Often in places with absent or captured (central/local) governments
 - ▶ Bandits collecting mineral taxation in Congo (Sanchez de la Sierra 2020)
 - ▶ Italian Mafia responsible for trash collection (some estimates say \$ 8.8B USD worth)
- Replaces state in some cases: Difficult to hold criminals accountable

Continuously contested question: Why?

- Why are some states unable to address criminal governance?
- Why do organized criminal groups engage in governance, not direct violence (or mix)?

Criminal governance is actually widespread

New Research Finds Nearly 80 Million Latin Americans Live Under Criminal Governance

January 17, 2023 (last updated on December 10, 2024)

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Findings suggest a correlation between gang governance and a positive perception of state governance.

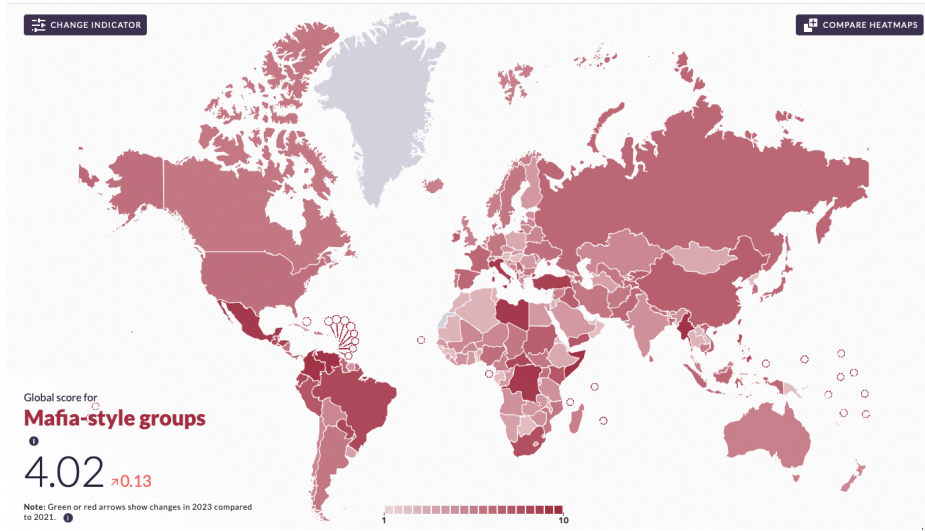
By Sarah Steimer

In nationally representative surveys across 18 Latin American countries, 13 percent of respondents say that local criminal groups provide order and reduce crime in their neighborhoods. This corresponds to 79.1 million residents, and researchers believe this likely underestimates the true number. In this [first systematic assessment of criminal governance across a region](#), published by SSRN and led by

faculty and doctoral students in the Department of Political Science at the University of Chicago, the research team also found that, when compared to the presence of non-governing gangs, criminal governance is correlated with greater state presence, more positive perceptions of state governance, and higher reported income.



Organized crime is not just limited to developing countries!



Source: Global Organized Crime Index (<https://ocindex.net/2023/heatmap/>)

Roadmap for today

1. Overview of the topic

2. Cause and consequences of individuals joining (organized) crime

- 2.1 Pinotti (2015): The Causes and Consequences of Organised Crime: Preliminary Evidence across Countries
- 2.2 Chimeli, Soares (2017): The Use of Violence in illegal Markets: Evidence from Mahogany Trade in Brazilian Amazon
- 2.3 Dix-Carneiro, Soares, Ulyssea (2018): Economic Shocks and Crime: Evidence from Brazilian Trade Liberalization [Student presentation]

3. Criminal governance: Relationship between formal state and organized criminal groups

- 3.1 Dell (2015): Trafficking Networks and the Mexican Drug War
- 3.2 Blattman, Duncan, Lessing, Tobón (2025): Gang Rule: Understanding and Countering Criminal Governance
- 3.3 Trejo, Ley (2018): Why Did Drug Cartels Go to War in Mexico? Subnational Party Alternation, the Breakdown of Criminal Protection, and the Onset of Large-Scale Violence [student presentation]

Pinotti (2015):
The Causes and Consequences of Organised
Crime: Preliminary Evidence across Countries

Q: What are the causes and consequences of organized crime?

Organized crime imposes serious consequences

- Violent activities can destroy physical and human capital
- Also increases uncertainties and riskiness → hurt long run investment and growth
- Yet, relationship among organized crime, economics, and institutions still uncertain

This paper provides cross-country empirical evidence on these relations

- Organized crime $\propto$ levels of per capita economic output $\Downarrow$
 - ▶ Two-way relation!!
- Politicians are more corrupt and/or vulnerable to attacks by organized crime
 - ▶ Suggest that organized crime → Weak institution
- Organized crime presence is positively correlated with natural resource endowment
 - ▶ But only in fragile states with weak legal capacity: Weak institution → organized crime

Measuring organized criminal group presence

By nature, measuring organized criminal group presence is tricky

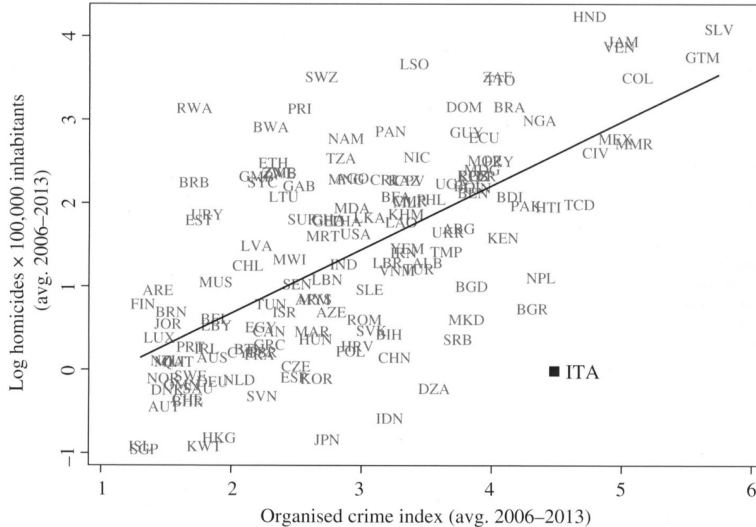
- Usually do not operate out in the open (and not always violent)
- When they do, they can pressure law enforcement to prevent detection

How this paper approaches this issue

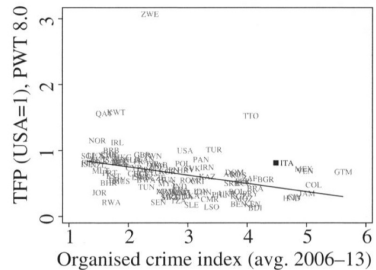
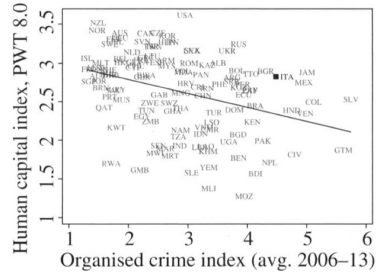
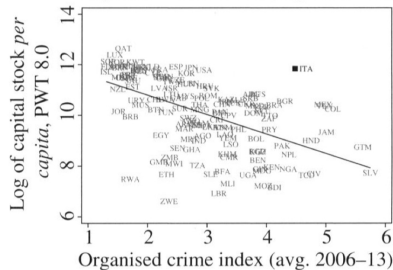
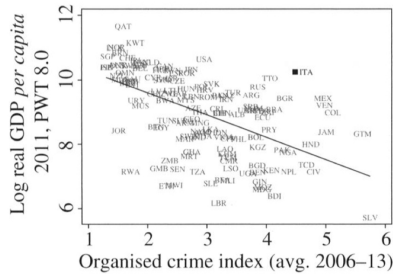
- Source: Executive Opinion Survey (World Economic Forum data) across 147 countries
 - ▶ 'In your country, to what extent does organized crime impose costs on business?'
 - ▶ Measured on 1 (least affected) to 7 (most affected) scale, averaged across '06-'13
- Economic performance from Penn World Table 8.0
- Political variables from World Bank Governance Indicators
- OLS to derive descriptive relations

$$Y_i = \alpha + \beta \text{Organized crime}_i + \gamma X_i + e_i$$

Organized crime correlates well with homicide rate



Organized crime negatively correlates with economic indicators



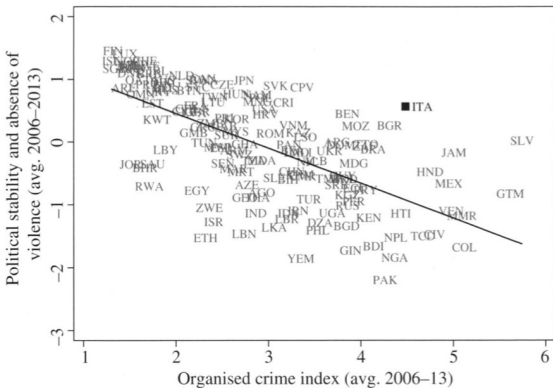
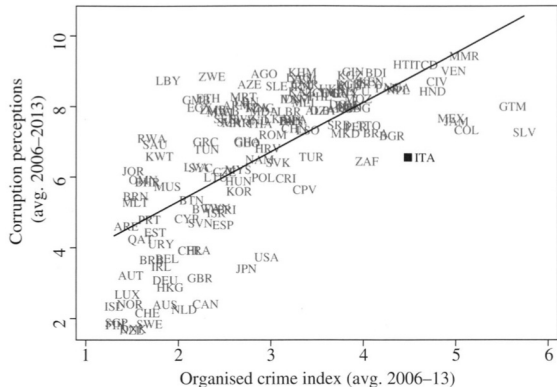
Organized crime negatively correlates with economic indicators

Table 2
Organised Crime and Economic Development

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Log of Real GDP <i>per capita</i> in 2011				<i>hc</i>	$\ln k$	<i>TFP</i>
Organised crime	-0.693*** (0.095)	-0.456*** (0.102)	-0.388*** (0.104)	-0.362*** (0.108)	-0.129*** (0.044)	-0.434*** (0.101)	-0.004 (0.026)
log Real GDP p.c. in 1950		0.626*** (0.091)	0.595*** (0.121)	0.589*** (0.118)	0.102 (0.087)	0.528*** (0.142)	0.229*** (0.038)
log Population in 1950		0.017 (0.062)	0.010 (0.105)	-0.016 (0.124)	-0.023 (0.056)	-0.016 (0.146)	-0.028 (0.034)
log Land Area		0.022 (0.052)	0.059 (0.085)	0.061 (0.094)	0.064 (0.040)	0.057 (0.110)	0.017 (0.028)
Primary exports, 1970			-0.800* (0.437)	-0.756 (0.464)	-0.391* (0.202)	-0.908* (0.539)	-0.155 (0.127)
Observations	137	110	95	95	88	95	76
Continent FE	No	Yes	Yes	Yes	Yes	Yes	Yes
Additional geo. controls	No	No	Yes	Yes	Yes	Yes	Yes
Coloniser FE	No	No	No	Yes	Yes	Yes	Yes
R ²	0.295	0.777	0.809	0.819	0.726	0.840	0.753

Notes. This Table reports the results of cross-country OLS regressions of the level and composition of aggregate output on organised crime, controlling for other country characteristics. The dependent variables are respectively, the log of GDP *per capita* at current PPPs (columns 1–4); an index number for human capital, $e^{\phi}(s)$, where s is the average number of years of schooling for the population aged 15 or older, and $\phi(s)$ are the returns to schooling at s years of education (column 5); the (log of) capital stock *per capita*, based on the perpetual inventory method (column 6); and total factor productivity relative to the US (column 7). All dependent variables are computed for year 2011, the source is the Penn World Tables. The measure of organised crime is based on the results of the Executive Opinion Survey averaged over the period 2006–13. The source of all other variables are Maddison (2007) and Nunn and Puga (2012). Robust standard errors are reported in parentheses; *, ** and *** denote coefficients significantly different from zero at the 90%, 95% and 99% confidence level respectively.

Organized crime negatively correlates with institutional quality



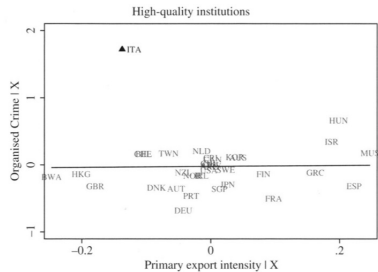
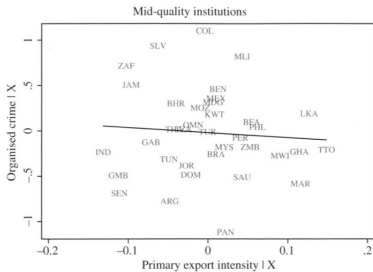
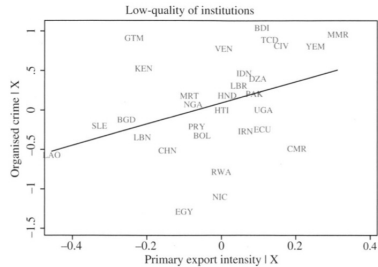
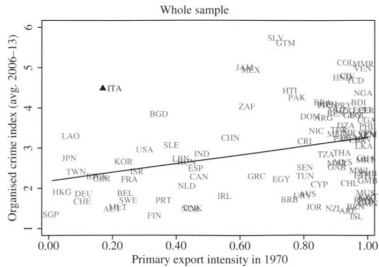
Organized crime negatively correlates with institutional quality

Table 3
Organised Crime, Corruption, and Political Stability

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Perceived corruption				Political stability & no violence			
Organised crime	1.403*** (0.143)	1.073*** (0.138)	0.969*** (0.146)	0.919*** (0.155)	-0.556*** (0.060)	-0.447*** (0.070)	-0.427*** (0.066)	-0.435*** (0.072)
log Real GDP p.c. in 1950		-0.746*** (0.194)	-0.588*** (0.209)	-0.606*** (0.205)		0.117 (0.079)	0.076 (0.079)	0.104 (0.074)
log Population in 1950		-0.310*** (0.103)	-0.134 (0.149)	-0.052 (0.153)		-0.058 (0.057)	-0.166*** (0.061)	-0.172*** (0.064)
log country land area		0.214** (0.096)	0.089 (0.117)	0.053 (0.128)		-0.010 (0.046)	0.100** (0.045)	0.077 (0.050)
Primary exports, 1970			2.224*** (0.603)	2.434*** (0.546)			-1.140*** (0.260)	-1.128*** (0.252)
Observations	147	118	100	100	147	118	100	100
Continent FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Additional geo. controls	No	No	Yes	Yes	No	No	Yes	Yes
Coloniser FE	No	No	No	Yes	No	No	No	Yes
R ²	0.472	0.758	0.823	0.841	0.412	0.593	0.735	0.788

Notes. This Table reports the results of cross-country OLS regressions of the quality of the political system on organised crime, controlling for other country characteristics. The dependent variables are the perceived level of corruption, as measured by Transparency International (columns 1–4) and an index of Political Stability and Absence of Violence, provided with the World Bank Governance Indicators (columns 5–8); both variables are averaged over the period 2006–13. The measure of organised crime is based on the results of the Executive Opinion Survey averaged over the period 2006–13. The source of all other variables is Maddison (2007) and Nunn and Puga (2012). Robust standard errors are reported in parentheses; *, ** and *** denote coefficients significantly different from zero at the 90%, 95% and 99% confidence level respectively.

Positively correlated with natural resource dependence in weak states



Discussion and takeaways

These findings highlight descriptive relations, but not causal

- Organized crime is hard to detect (measurement error)
- Organized crime is a result of economic and political conditions (reverse causality)

Many modern research on the topic tries to unearth causal relationship

- Leverage sudden economic shock: Economic condition → Organized crime
- Leverage political events (e.g. assassinations): Political event → Organized crime
- IV method (e.g. road networks, agriculture): Organized crime → Economics and politics
- In a rare instance, some people design an randomized experiments

Chimeli, Soares (2017):
The Use of Violence in illegal Markets: Evidence
from Mahogany Trade in Brazilian Amazon

Q: How does illegalization of a market cause violence?

In illegal markets, state presence is limited

- No formal means to enforce contracts and guarantee property rights
- Violence potentially is used as arbitration device
- But causal link between market illegality and violence is limited
 - ▶ Extremely rare, unable to do randomized experiment

This paper uses illegalization of mahogany trade in Brazil to answer this question

- Leverage variation in homicide rates across areas with different mahogany shares
- Homicide rates $\uparrow$ in high mahogany growth areas
- This is amplified in states with high mahogany export shares
- Homicides subsided with federal effort to reduce illegal mahogany trades itself

Mahogany trades in Brazil and violence

Brazil as a Mahogany exporter

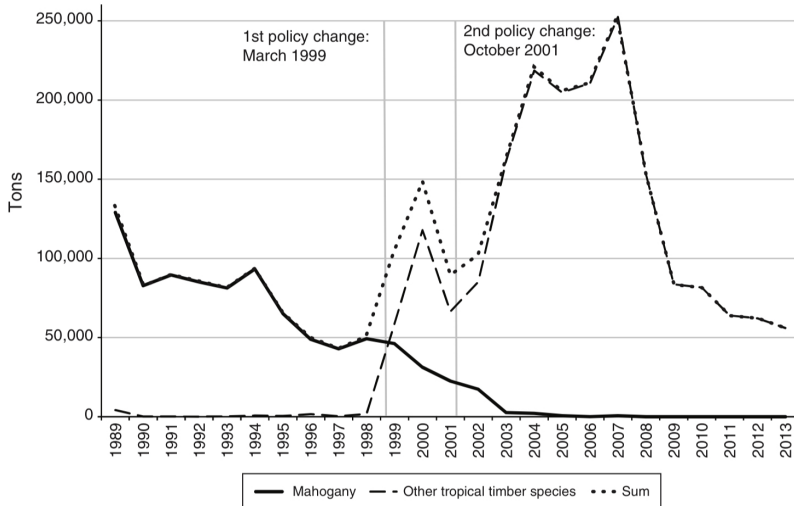
- One of the largest exporters before prohibition in 2001
- Illegalization: Limited issuing of licenses in 1999, and introduced export quota at 2001
- Illegalization of mahogany markets since
 - ▶ Traded as 'other timber species'
 - ▶ Trade picked up until active monitoring and enforcement came in at late in 2000s

Violence and illegal mahogany markets

- No formal credit markets or contracts available → resort to violence
- Use intimidation and threats on indigenous tribe leaders to steal Mahogany
- Also use illegal weapons in the process
- These were sometimes carried out by organized criminal groups in the area

Mahogany trades persisted under 'others'

Panel B. Yearly exports of mahogany and other tropical timber species



Data and empirical strategy

Data sources

- Mahogany presence: Brazilian Secretariat on International Trade
- Violence: Ministry of Health keeps homicide rates at municipality level
- Also control for mortality, land conflicts, total areas planted, agricultural GDP etc

Diff-in-diff leverages variation in mahogany and violence across time and municipalities

$$Y_{it} = \alpha + \beta_1 D_{['99 \leq t \leq '01]} \times M_i + \beta_2 D_{['02 \leq t \leq '08]} \times M_i + \beta_3 D_{[t \geq '09]} \times M_i + \gamma Z_{it} + \phi_i + \mu_{st} + e_{it}$$

- Breaks down into 3 periods: Different modes of illegalization and enforcement
- Outcome variables: Primarily homicide rates

Illegalization increases violence

Variables	Municipalities in states with mahogany occurrence					
	(1)	(2)	(3)	Treatments interacted with linear trends (4)	Triple-diff:	
					State percent in exp. before 1999 (5)	Suspect. state exp. after 1999 (6)
<i>treat 1999</i>	4.520 [2.627]	8.078 [2.841]	5.946 [2.031]	5.669 [2.586]	17.13 [6.078]	0.0994 [0.0370]
<i>treat 1999 × trend</i>				2.409 [1.398]		
<i>treat 2002</i>	7.034 [3.491]	15.03 [3.633]	12.68 [3.650]	15.11 [3.321]	31.13 [7.517]	0.139 [0.0303]
<i>treat 2002 × trend</i>				-0.0275 [0.556]		
<i>treat 2009</i>	-1.478 [4.521]	9.514 [4.207]	10.81 [4.173]	14.29 [5.369]	22.52 [9.228]	0.371 [0.146]
<i>treat 2009 × trend</i>				-2.387 [1.023]		
State FE × year FE		X	X	X	X	X
Baseline charact. × year FE			X			
Observations	11,932	11,932	11,533	11,932	11,932	11,932
R^2	0.645	0.709	0.772	0.710	0.712	0.711

Particularly in a state with very high Mahogany exports

Variables	Municipalities in Pará			Municipalities in states with mahogany occurrence excluding Pará			
	(1)	(2)	Treatments interacted with linear trends (3)	(4)	(5)	(6)	Treatments interacted with linear trends (7)
<i>treat 1999</i>	12.68 [4.458]	11.00 [3.214]	9.642 [4.154]	0.346 [3.457]	1.600 [2.206]	2.734 [2.001]	0.0811 [1.900]
<i>treat 1999 × trend</i>			3.041 [2.179]				1.519 [1.330]
<i>treat 2002</i>	22.59 [5.520]	18.71 [4.966]	20.93 [4.963]	-1.993 [3.077]	4.392 [2.241]	4.466 [1.694]	6.924 [2.809]
<i>treat 2002 × trend</i>			0.553 [0.886]				-0.844 [0.452]
<i>treat 2009</i>	16.49 [6.771]	20.89 [6.959]	25.60 [8.555]	-12.48 [2.876]	-0.299 [2.796]	2.819 [2.604]	-1.628 [2.278]
<i>treat 2009 × trend</i>			-4.555 [1.517]				0.665 [0.716]
State FE × year FE					X	X	X
Baseline charact. × year FE		X				X	
Observations	2,432	2,033	2,432	9,500	9,500	9,500	9,500
R^2	0.731	0.864	0.735	0.651	0.694	0.745	0.694

Victims usually prime aged male

Variables	All	Male homicides		Male homicides, ages 15–39			Deaths from land conflicts	
	1996–2013	All ages	Ages 15–39	Single	Death occurred outside the home	Homicide by firearm	Rate (per 100,000)	Occurrence dummy
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>treat 1999</i>	11.36 [3.346]	22.59 [6.041]	37.37 [11.63]	22.35 [8.771]	32.87 [9.858]	26.20 [7.650]	−0.00592 [0.00376]	−0.0260 [0.0384]
<i>treat 2002</i>	21.27 [5.269]	38.17 [10.32]	67.87 [19.30]	52.66 [17.57]	63.23 [17.75]	41.01 [16.35]	−0.000448 [0.00501]	0.00145 [0.0806]
<i>treat 2009</i>	15.16 [6.885]	25.40 [13.86]	30.33 [26.39]	26.12 [23.80]	26.56 [25.86]	2.028 [27.53]		
1996 level of dependent variable	14.29	24.09	35.72	34.09	25.33	20.78	0.01	0.11
Proportional effect in 2002–2008	148%	158%	190%	154%	250%	197%	−3%	1%
Observations	2,304	2,304	2,304	2,304	2,304	2,304	1,664	1,664
R^2	0.753	0.754	0.733	0.734	0.734	0.694	0.167	0.356

- Suggests that deaths are due to criminal dispute over domestic violence

Discussion and takeaways

Illegality of markets can increase violence

- These are not explained by socioeconomic conditions
- Monitoring and enforcement $\rightarrow$ Illegal market $\Downarrow \rightarrow$ Violence $\Downarrow$

What cautionary tale does it provide?

- In absence of adequate monitoring and enforcement, addressing unwanted externalities with overly restrictive regulations may increase social loss
 - ▶ When markets become illegal, violence replaces formal credit markets and contracts
- But how to establish capable monitoring?
 - ▶ Corruption can make monitoring difficult (police implicitly cooperating with bandits)

Dix-Carneiro, Soares, Ulyssea (2018):
Economic Shocks and Crime: Evidence from
Brazilian Trade Liberalization [Student
presentation]

Dell (2015):
Trafficking Networks and the
Mexican Drug War

Q: How do organized criminal groups respond to crackdowns?

Many countries engage in counterinsurgency strategies to drive out organized crime

- Mexico: Spent \$9 billion USD per year ($\simeq$ Social development expenses)
- Causal evidence of crackdowns is rare
 - ▶ Selection bias: State chooses where crackdown takes place deliberately
 - ▶ Reverse causality: Governments choose places likely to expect high violence

This paper uses plausibly exogenous variation in crackdowns

- Closely elected mayors of the party spearheading crackdowns (PAN)
- Places with close elections are less likely to be different on other dimensions
- Driven by drug traffickers killing each other (incumbent gang vs new gangs)
- PAN victories divert trafficking route, increase violence on alternate routes

Background information

Drug Trafficking in Mexico

- Large industry: \$14-48 billion USD (mostly to US)
- Drug trafficking organizations (DTOs) present in 68% of Mexican municipalities
- Number of operating DTOs increasing since 2006 (also engage in other illicit activities)

Crackdowns on the DTOs

- Felipe Calderón (PAN): Made crackdowns a centerpiece of his administration
 - ▶ Mexico was previously a one-party (PRI) country, which took passive stance on DTOs
 - ▶ Federal police/military arrest high-rank traffickers and seize drugs
- Mayors: Name police officers who can assist federal government crackdowns
 - ▶ PAN mayors were actively supporting this
 - ▶ Also more likely to request federal help compared to non-PAN mayors

Data and empirical strategy

Data sources

- Official govt data on drug trades, DTO presence and related crimes (confidential)
- Election data from state electoral committees (available right now)

Effects of close PAN victory (i.e. increased crackdown)

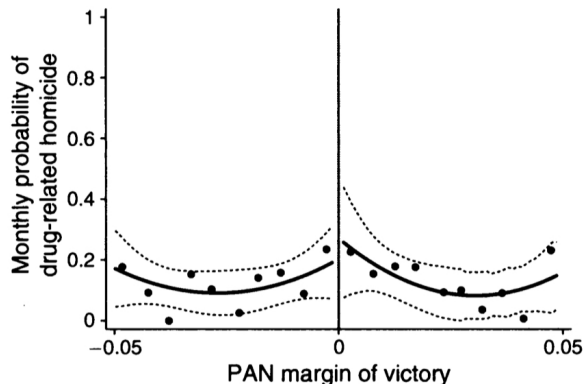
- Regression discontinuity leveraging close margin of victory for PAN

$$Y_m = \alpha_0 + \alpha_1 \text{PAN}_m + \alpha_2 \text{PAN}_m \times f(\text{margin}_m) + \alpha_3 (1 - \text{PAN}_m) \times f(\text{margin}_m) + e_m$$

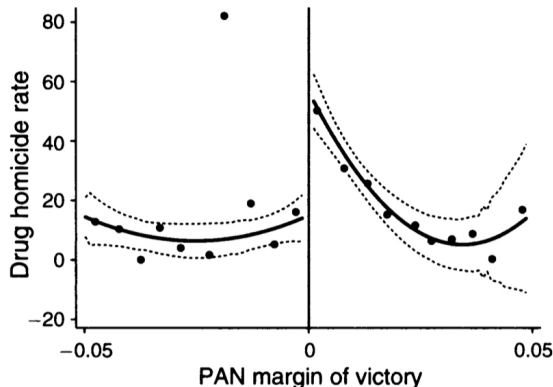
- Close election of PAN mayors vs Close loss of PAN mayors
- Other characteristics equal: Confirms that no other characteristics differ (Table 1)

Existing drug-related violence intensifies

Panel A. Post-inauguration (extensive margin)



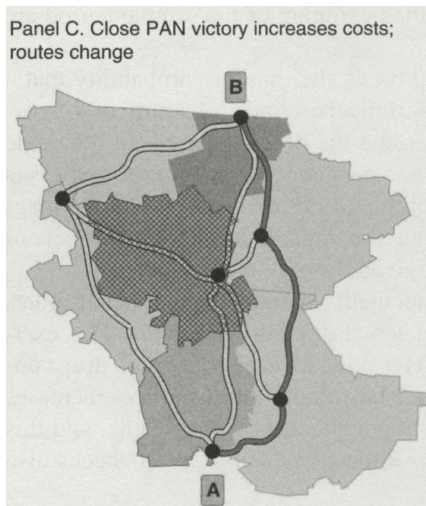
Panel B. Post-inauguration (homicide rate)



- Driven by rival gangs attacking incumbent gangs weakened by enforcements
- Does not happen pre-election or pre-inauguration after elections (Rest of Fig 3)

Do DTO responses have spillovers?

Logic behind spillovers to other regions



- Close PAN victories raise costs of trafficking
- This incentivizes DTOS to shift paths
- Leading to increase in drug-related violence elsewhere

Indeed, violence do spillover

	Dependent variable: Drug trade-related homicide							
	Full sample				Limited sample			
	Dummy (1)	Rate (2)	Dummy (3)	Rate (4)	Dummy (5)	Rate (6)	Dummy (7)	Rate (8)
<i>Panel A. Shortest paths</i>								
Predicted routes dummy	0.014*** (0.005)	1.163 (1.251)			0.006 (0.006)	-0.702 (1.199)		
One route			0.017** (0.007)	-6.180 (3.779)			0.014 (0.010)	-5.530 (3.449)
More than one route			0.011 (0.008)	10.314** (4.746)			-0.000 (0.010)	6.164 (4.511)
<i>Panel B. Model with congestion costs</i>								
Predicted routes dummy	0.017*** (0.005)	1.870** (0.811)			0.019*** (0.006)	1.831** (0.930)		
One route			0.010 (0.006)	2.188 (1.631)			0.010 (0.007)	1.399 (0.943)
More than one route			0.020*** (0.006)	1.746* (1.058)			0.023*** (0.007)	2.019* (1.034)
Municipalities	1,869	1,869	1,869	1,869	1,562	1,562	1,562	1,562
Observations	69,153	69,153	69,153	69,153	57,794	57,794	57,794	57,794

Takeaways and discussions

Crackdowns induce direct and spillover costs

- Violence in PAN municipalities $\uparrow$ ($\because$ rival gangs kill weakened incumbent gangs)
- Violence spills over to nearby municipalities ($\because$ reroute to find cheaper transport costs)
- Thus, DTOs respond to changing environments, offsetting effectiveness of crackdowns

So what does it inform about crackdown policies?

- Emphasize ways to deter violence at its roots
 - ▶ Other ways to disincentivize violence (very tricky question)
- Focusing on high-rank arrests may open other gangs to take over
 - ▶ Especially in places where inter-cartel competition is fierce (Mexico)
 - ▶ End up having criminal governance by rival gangs

Blattman, Duncan, Lessing, Tobón (2025):
: Gang Rule: Understanding and Countering
Criminal Governance

Q: Can state and criminal governance coexist? How?

Classical explanation focuses on the *monopolistic* governance of state or criminal groups

- Fill in the vacuum: Criminal groups fill in the absence of the state
- Competition: State exerts influence by driving out criminal groups (and vice versa)
- But there are cases where *duopolistic* rule exists (Italy, South Africa, Latam etc.)
 - ▶ Criminal groups rule civilians and coexist with state governance

This paper identifies how duopolistic governance becomes possible

- Gangs govern civilians to protect illegal revenues (drug dealing)
- In settings where such gains are high, state presence $\uparrow \Rightarrow$ Criminal rule $\uparrow$
- Thus, state and criminal rule becomes complements, not substitutes
- Leverages exogenous boundary changes that changes the distance to closest state HQ
- Distance to HQ $\downarrow \rightarrow$ state responsiveness and criminal presence $\uparrow$

Context and framework

Context: Medellín, Colombia

- Has well-organized bureaucracy and large fiscal revenues
- Security and legal services provided at sub-city level
- Poorer blocks have *combos*: small local gangs providing security and drug trade

Why do gangs provide security services? (interview with gang leaders)

- Providing security reduce need for police presence
- Expected residents to be loyal (i.e. cooperate with gangs over state)

How can state-gangs be complementary?

- Classic: When state fails to meet demand for governance, gangs enter
- Twist: Simultaneous governance protects gangs illicit business from state repression
 - ▶ Illicit profit ↑ & gang-state services differ: Gang incentive to govern areas close to state ↑
 - ▶ Low profit or state already dominant: Incentive to govern ↓

Data and empirical strategy

Data sources

- Interviews with the criminal group leaders to understand motives and combo presence
- Surveys on residents and businesses to gather data on governance and perceptions

Geographic regression discontinuity that leverages 1987 reform

- Analysis at block-pairs
- New borders in Medellín (6 sub-area $\rightarrow$ 16) divided a subregion into 2-3 smaller units
- Changes the nearest security and justice service location for some blocks
- Regress the following equation: Δd_{ij} is the treatment (i, j : block pairs, b : border)

$$\Delta Y_{ijb} = \alpha_b + \beta \Delta d_{ij} + \theta \Delta X_{ij} + \lambda B_{ij} + e_{ijb}$$

- Confirm that other characteristics do not change with the new border introduction

Description of the regression discontinuity

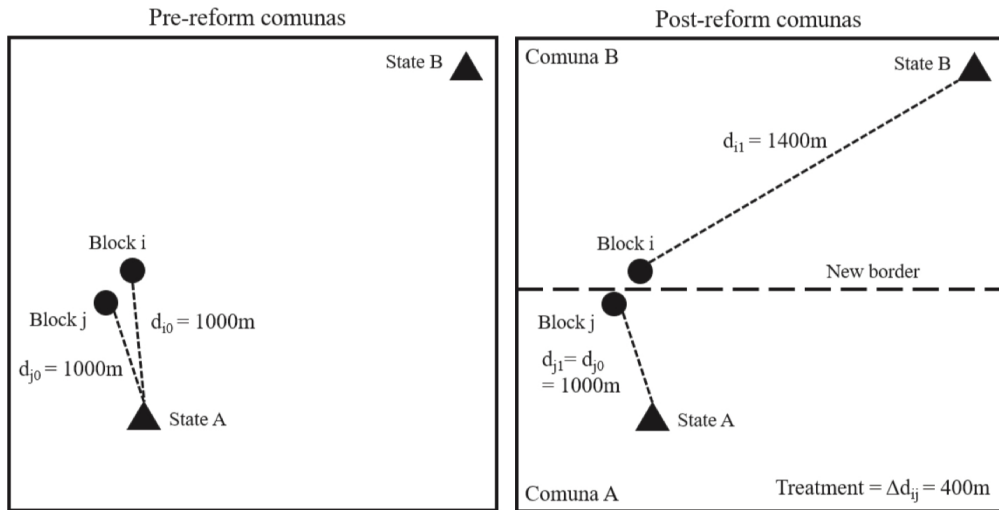


FIGURE 1
Stylized illustration of the natural experiment

State and combo presence increases as distance to HQ decreases

TABLE 3
Impacts on governance and legitimacy of being 100 metres closer to the state

Effect on ΔY	Subsample Mean (SD) (1)	Effect of moving 100 m closer to state Estimate (SE) [p-value] (2)	Median distance: 400m Median change as % of sample mean Estimate (3)
Governance ($N = 570,563$)			
Δ State Governance Index (0–1)	0.41 (0.19)	0.011 (0.007) [0.186]	10.8
Δ Combo Governance Index (0–1)	0.32 (0.22)	0.019** (0.007) [0.027]	24.1
Accessibility ($N = 429,384$)			
Δ State Accessibility Index (0–1)	0.42 (0.16)	0.012* (0.006) [0.050]	11.6
Δ Combo Accessibility Index (0–1)	0.54 (0.23)	0.023** (0.008) [0.024]	17.7
Legitimacy ($N = 431,425$)			
Δ State Legitimacy Index (0–1)	0.58 (0.14)	0.001 (0.008) [0.883]	0.9
Δ Combo Legitimacy Index (0–1)	0.42 (0.21)	0.005 (0.014) [0.749]	4.8

Combo response strong in areas with high drug rents

Regression augmented with measure of illicit rent π_{ij}

$$\Delta Y_{ijb} = \alpha_b + \beta \Delta d_{ij} + \gamma \pi_{ij} + \delta \Delta d_{ij} \times \pi_{ij} + \theta \Delta X_{ij} + \lambda B_{ij} + e_{ijb}$$

TABLE 4

Heterogeneity: Impacts on governance and legitimacy of being 100 metres closer to the historical state, by predicted proximity to major drug plazas

	Sample Mean (SD) (1)	Governance (Δ)		Accessibility (Δ)		Legitimacy (Δ)	
		State Estimate (SE) [p-value]	Combo Estimate (SE) [p-value]	State Estimate (SE) [p-value]	Combo Estimate (SE) [p-value]	State Estimate (SE) [p-value]	Combo Estimate (SE) [p-value]
		(2)	(3)	(4)	(5)	(6)	(7)
Effect of moving 100 m closer to state	−4.966 (4.378)	0.011 (0.007) [0.114]	0.017* (0.007) [0.074]	0.009 (0.006) [0.108]	0.023** (0.010) [0.023]	0.002 (0.007) [0.787]	0.004 (0.015) [0.850]
100 m closer to state X Predicted drug profitability (std.)	−2.016 (5.478)	0.006 (0.004) [0.158]	0.010** (0.004) [0.039]	−0.002 (0.003) [0.46]	−0.004 (0.004) [0.90]	−0.010** (0.004) [0.044]	−0.011** (0.004) [0.024]
Controls and fixed point running var		✓	✓	✓	✓	✓	✓
Bi-directional Border FE		✓	✓	✓	✓	✓	✓
N		563	555	429	384	429	423

Presence of criminal groups does not increase perceived legitimacy

Legitimacy: combination of trust, fairness, and satisfaction

- State and combo legitimacy does not increase (two slides before)
- In high drug rent areas, closer $\Rightarrow$ legitimacy for both state and combos $\downarrow$ (slide before)
- Qualitative interview results match: Residents unhappy with influx of 'undesirable' ppl
- No significant economic development for most measures...

Why did this happen?

- Gangs overestimating how loyal citizens were to them
- Gangs entering areas to offset loyalty penalty coming from drug sales
- Yet, providing security could reduce police patrol (so motives to enter exist)

Takeaways and discussions

State and criminal presence can be complementary

- Particularly in areas where returns from illegal trades are high
- Could explain converse case: Low illicit returns $\Rightarrow$ Gang extortions rise

What is to be learned from this? (ongoing question)

- To cut criminal presence, need to attack root causes that incentivize them
 - ▶ Tougher than said, what are the criminal groups after to begin with?
 - ▶ Criminal groups can strategically move to retain these benefits (see any drug wars)
- Can the findings be generalized, even in places with high extent of illicit activities?
 - ▶ Active state vs. organized criminal conflict in such places too (see Mexico)

Trejo, Ley (2018):

Why Did Drug Cartels Go to War in Mexico?
Subnational Party Alternation, the Breakdown
of Criminal Protection, and the Onset of
Large-Scale Violence [student presentation]

Takeaways and remaining questions

Many individuals join organized crime, some of which evolve to criminal governance

- Particularly in economically or institutionally less developed areas
- Illegalization of the markets potentially gives organized crime an opportunity
- Some of them are on verge to compete governments out
 - ▶ Either strategically coexisting in and duopoly
 - ▶ Or adjusting their survival strategy accordingly
 - ▶ Sometimes outright fighting formal state

Unanswered questions

- How do you effectively curb incentives to join crime?
 - ▶ Monitoring requires transparent and accountable government: Not always the case
 - ▶ Hardline stance can go wrong (Mexico) but it can deliver a strong message
- What can be done to address criminal governance?
 - ▶ Organized crime not accountable to the populace: How to break up these groups?
 - ▶ Minimizing state absence/capture: Organized crime presence limited in capable states